

UNIVERSITÀ
DEGLI STUDI
DI MILANO

Calibration Of Probabilities

Luca Rigamonti 08920A

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1 Introduction

The project try to find out whether the probabilities produced by a classifier are reliable. A model is considered calibrated if, among all the cases to which it assigns a probability of 0.8, approximately 80% are truly positive. A model can have excellent accuracy while still providing probabilities that are not very truthful.

In this project, I used two real-world datasets, which were divided into a training set, a validation set (used for learning calibration), and a test set. To correct the probabilities, two methods are implemented from scratch on the validation set: Platt Scaling and Isotonic Regression .
Performance on the test set is measured using three metrics (Accuracy, Brier Score, and Log-Loss) that were implementad from scratch.

2 Data

The first dataset is "Wine Quality," taken from the Vinho Verde collection from Portugal, which is available in the UCI repository.t includes eleven chemical-physical variables measured in a laboratory—such as acidity level, residual sugars, pH, and alcohol content—that describe the objective properties of the wine. The objective involves estimating quality based on sensory tests, with scores ranging from 0 to 10; for the purposes of this study, the target variable is transformed into a binary classification by introducing a threshold, labeling each sample as "good" (≥ 6) or otherwise.

```
df_wine.sample(10, random_state=55)
```

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol	quality	wine_type
4083	6.2	0.30	0.32	1.2	0.052	32.0	185.0	0.99266	3.28	0.44	10.100000	5	1
1898	7.2	0.31	0.41	8.6	0.053	15.0	89.0	0.99760	3.29	0.64	9.900000	6	1
4188	5.3	0.33	0.30	1.2	0.048	25.0	119.0	0.99045	3.32	0.62	11.300000	6	1
3117	6.5	0.23	0.39	1.9	0.036	41.0	98.0	0.99000	3.19	0.43	11.900000	7	1
4656	6.0	0.29	0.41	10.8	0.048	55.0	149.0	0.99370	3.09	0.59	10.966667	7	1
4382	7.0	0.24	0.26	1.7	0.041	31.0	110.0	0.99142	3.20	0.53	11.000000	6	1
5955	7.6	0.42	0.25	3.9	0.104	28.0	90.0	0.99784	3.15	0.57	9.100000	5	0
849	6.0	0.20	0.24	1.8	0.030	30.0	105.0	0.99090	3.31	0.47	11.500000	6	1
1618	6.4	0.37	0.49	13.3	0.045	53.0	243.0	0.99820	3.14	0.48	8.500000	6	1
1325	7.7	0.32	0.48	2.3	0.040	28.0	114.0	0.99110	3.20	0.52	12.800000	7	1

The second dataset is the Breast Cancer Wisconsin (Diagnostic), a biomedical dataset related to cancer diagnosis. The dataset consists of 30 variables related to tumor cells obtained with fine-needle aspiration . The predictive task is a classic binary classification, distinguishing if a cell mass is benign or malignant.

```
df_cancer.frame.sample(10, random_state=55)
```

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	worst texture	worst perimeter	worst area	worst smoothness	worst compactness	worst concavity	worst concave points	worst symmetry	worst fractal dimension	target
446	17.75	28.03	117.30	981.6	0.09997	0.13140	0.16980	0.08293	0.1713	0.05916	...	38.54	145.40	1437.0	0.1401	0.3762	0.63990	0.19700	0.2972	0.09075	0
523	13.71	18.68	88.73	571.0	0.09916	0.10700	0.05385	0.03783	0.1714	0.06843	...	25.63	99.43	701.9	0.1425	0.2566	0.19350	0.12040	0.2849	0.09031	1
530	11.75	17.56	75.89	422.9	0.10730	0.09713	0.05282	0.04440	0.1598	0.06677	...	27.98	88.52	552.3	0.1349	0.1854	0.13660	0.10100	0.2478	0.07757	1
353	15.08	25.74	98.00	716.6	0.10240	0.09769	0.12350	0.06553	0.1647	0.06464	...	33.22	121.20	1050.0	0.1660	0.2356	0.40290	0.15260	0.2654	0.09438	0
173	11.08	14.71	70.21	372.7	0.10060	0.05743	0.02363	0.02583	0.1566	0.06669	...	16.82	72.01	396.5	0.1216	0.0824	0.03938	0.04306	0.1902	0.07313	1
30	18.63	25.11	124.80	1088.0	0.10640	0.18870	0.23190	0.12440	0.2183	0.06197	...	34.01	160.50	1670.0	0.1491	0.4257	0.61330	0.18480	0.3444	0.09782	0
458	13.00	25.13	82.61	520.2	0.08369	0.05073	0.01206	0.01762	0.1667	0.05449	...	31.88	91.06	628.5	0.1218	0.1093	0.04462	0.05921	0.2306	0.06291	1
34	16.13	17.88	107.00	807.2	0.10400	0.15590	0.13540	0.07752	0.1998	0.06515	...	27.26	132.70	1261.0	0.1446	0.5804	0.52740	0.18640	0.4270	0.12330	0
329	16.26	21.88	107.50	826.8	0.11650	0.12830	0.17990	0.07981	0.1869	0.06532	...	25.21	113.70	975.2	0.1426	0.2116	0.33440	0.10470	0.2736	0.07953	0
408	17.99	20.66	117.80	991.7	0.10360	0.13040	0.12010	0.08824	0.1992	0.06069	...	25.41	138.10	1349.0	0.1482	0.3735	0.33010	0.19740	0.3060	0.08503	0

10 rows × 31 columns

3 Methodology

3.1 Data preparation

To begin with, the datasets were both turned into binary classification problems. In paricular, For Wine Quality, the red and white wines were merged into a single dataset (the wine type is kept as a feature) and the quality score was semplified;

- A wine is labeled positive ("good") if its quality is at least 6.
- For Breast Cancer, the original binary target was kept (1 = benign, 0 = malignant).

Both dataset were splitted into train (60%), validation (20%) and test (20%) sets. The calibration set is independent from the training set because calibrating on the data used to train the model would give biased estimates. The test set was used only for the final evaluation.

In this project, I trained logistic regression (with max iter=5000) and a Random forest (with 200 trees) on the training set, using the off-the-shelf scikit-learn implementations.

3.2 Calibration

After that, I implemented 2 calibration methods: item **Platt scaling.** The score is the logit of the probability provided by the model, and the calibrated probability is given by

$$p = \frac{1}{1 + \exp(A \cdot f + B)}.$$

The parameters A and B were determined by minimizing the cross-entropy using gradient descent. As in the reference paper, I used "smoothed targets" in order to reducce overfitting.

Isotonic Regression I used a increasing step function using the Pool Adjacent Violators (PAV) algorithm.

4 Evaluation

4.1 Wine Dataset

I begin by analyzing the results on the wine dataset.

Model	Accuracy	Brier Score	Log-loss
Basic LR	0.7454	0.1771	0.5309
LR + Platt	0.7438	0.1773	0.5321
LR + Isotonic	0.7415	0.1773	0.7341
Basic RF	0.8131	0.1320	0.4122
RF + Platt	0.8154	0.1306	0.4073
RF + Isotonic	0.8154	0.1326	0.4134

4.2 Logistic Regression

Platt Scaling It is practically identical to the basic version (Brier 0.1771 –¿ 0.1773, log-loss 0.5309 –¿0.5321). The LR model was already calibrated, so Platt did not correct anything.
Isotonic: the Brier score remains unchanged (0.1773), but the log-loss spikes (0.5309 –¿ 0.7341). This is because Brier score and log-loss penalize confident errors in very different ways: the Brier score reaches a maximum of 1 for a prediction that is both certain and incorrect, while log-loss can reach approximately 34.

Isotonic Regression Isotonic regression produces a step function that can have on values of exactly 0 or 1 at the extremes. If a wine in the test set falls into a step at 0 (or 1) but is actually the opposite, the log-loss increases significantly, while the Brier scoredid not change.

4.3 Random Forest

Platt Scaling Platt scaling performs best: Brier score improves from 0.1320 to 0.1306, log-loss from 0.4122 to 0.4073, and accuracy also sees a slight increase (from 0.8131 to 0.8154).

Isotonic Regression Isotonic regression results are nearly identical to the basic output—though slightly worse (Brier 0.1326, log-loss 0.4134). With approximately 1,300 calibration data points, it doesn't perform poorly, but it has no significant improvement.

Model	Accuracy	Brier Score	Log-loss
Basic LR	0.9565	0.0296	0.1015
LR + Platt	0.9565	0.0305	0.1118
LR + Isotonic	0.9565	0.0261	0.6510
Basic RF	0.9739	0.0371	0.1368
RF + Platt	0.9739	0.0379	0.1423
RF + Isotonic	0.9217	0.0531	0.4366

4.4 Breast Cancer

In this case, calibration doesn't help and, in one case, it actually has an adverse impact. But, A preliminary note; the test set consists of approximately 115 patients, meaning each patient accounts for 0.87 percentage points of accuracy: an accuracy of 0.9565 corresponds to 110 correct out of 115. So small differences (such as those between Platt scaling and basic scores) are probably just noise.

4.5 Logistic Regression

Platt Scaling slightly worse than the raw output (Brier 0.0296 –¿ 0.0305, log-loss 0.1015 –¿ 0.1118). Accuracy remains identical.

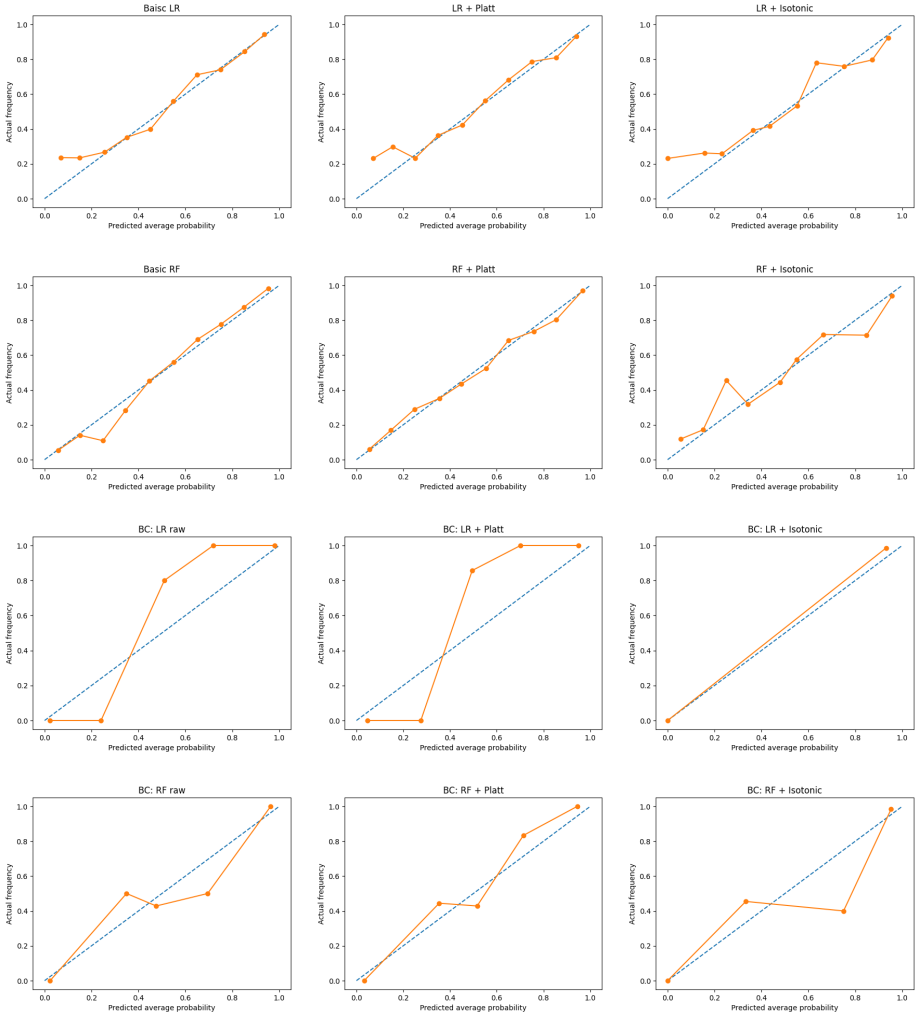
Isotonic Regression The Brier score improves slightly (0.0261), but the log-loss rises from 0.10 to 0.65.

4.6 Random Forest

Platt Scaling slightly worse again (0.0371 –¿ 0.0379; 0.1368 –¿0.1423), with accuracy remaining the same.

Isotonic Regression It makes everything worse: accuracy drops from 0.9739 to 0.9217 (6 additional misclassified patients), Brier score goes from 0.0371 to 0.0531, and log-loss from 0.1368 to 0.4366. So, This is a case of the overfitting described also in the paper: with too few calibration points, the step function adapts to the noise in the validation set and generalizes poorly.

4.7 Charts


Baisc LR
LR + Platt
LR + Isotonic
Basic RF
RF + Platt
RF + Isotonic
BC: LR raw
BC: LR + Platt
BC: LR + Isotonic
BC: RF raw
BC: RF + Platt
BC: RF + Isotonic
Actual frequency
Predicted average probability

5 Final comparison

- First of all, comparing the result I can say that the size of the calibration set is important. this is the key finding, In fact, Applying isotonic regression to the Wine dataset using a Random Forest has almost no negative impact, while the same combination makes things significantly worse for the Breast Cancer dataset. The main difference between the two cases is that the validation set in the latter contains a lot fewer data points. This aligns with what the paper "Predicting Good Probabilities with Supervised Learning": isotonic regression requires approximately 1,000 data points or more; otherwise, it leads to overfitting.

- Platt is the only "safe" method, but it only helps where there is something to correct. For the Wine dataset, it provides a slight improvement to the RF [Random Forest] model. For the Breast Cancer dataset, it is slightly Adverse.

- For the Breast Cancer dataset, there is little to correct. The reliability diagrams contain few useful points in the middle ranges. In the case of the Wine dataset, however, the probabilities are distributed across the entire range; so, miscalibration is more evident and can be corrected.

- For LR + Isotonic on the Breast Cancer dataset, the Brier score improves (0.0296 –¿0.0261), while the log-loss becomes six times worse. Log-loss penalizes confident errors, and Isotonic calibration tends to produce extreme probabilities. So, It depends on the metric how bad the calibration appears.